AI and Visualisation: Phase 1
Team Kangaroos

K12212960, K12209301, K12503533, K12545112

Fashion-mnist dataset

Fashion-MNIST is a machine learning dataset created by Zalando Research (introduced by Han Xiao, Kashif Rasul, and Roland Vollgraf in 2017) as a direct drop-in replacement for the classic handwritten-digit MNIST dataset

Dataset: Fashion MNIST Dataset

Size: 70k images

10k for testing

60K for training

With 10 classes

Image Format: 28 × 28 pixels per image (784 features), with pixel values ranging from 0 to 255

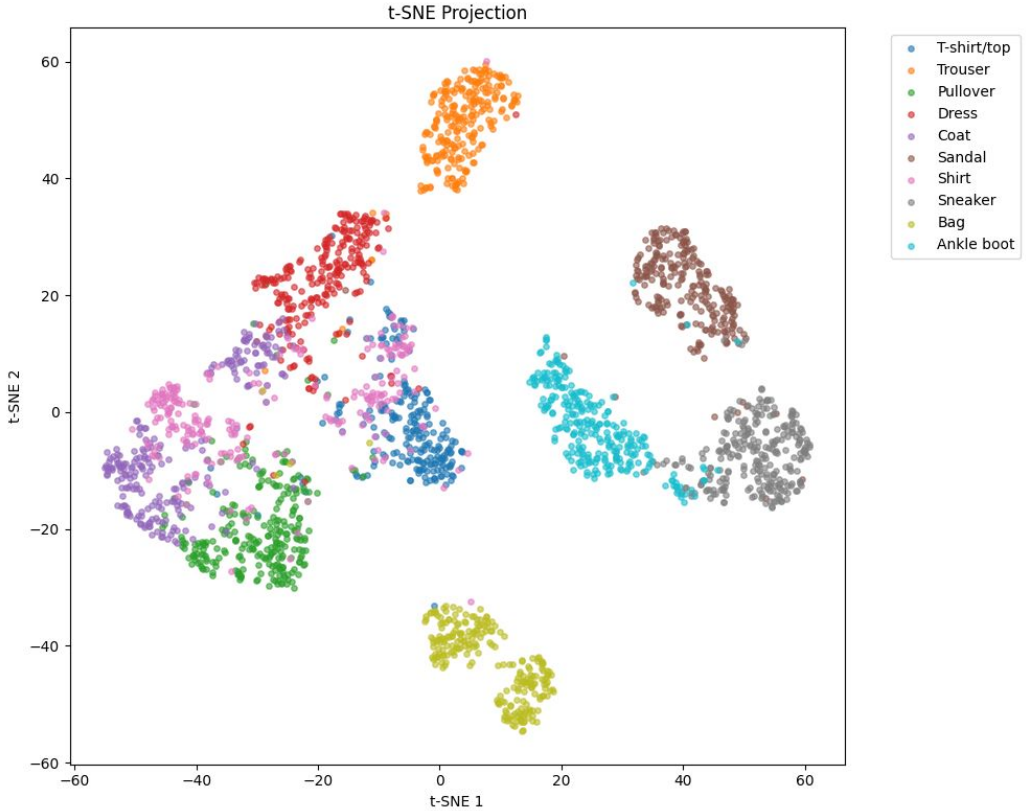
Model: ResNet 18



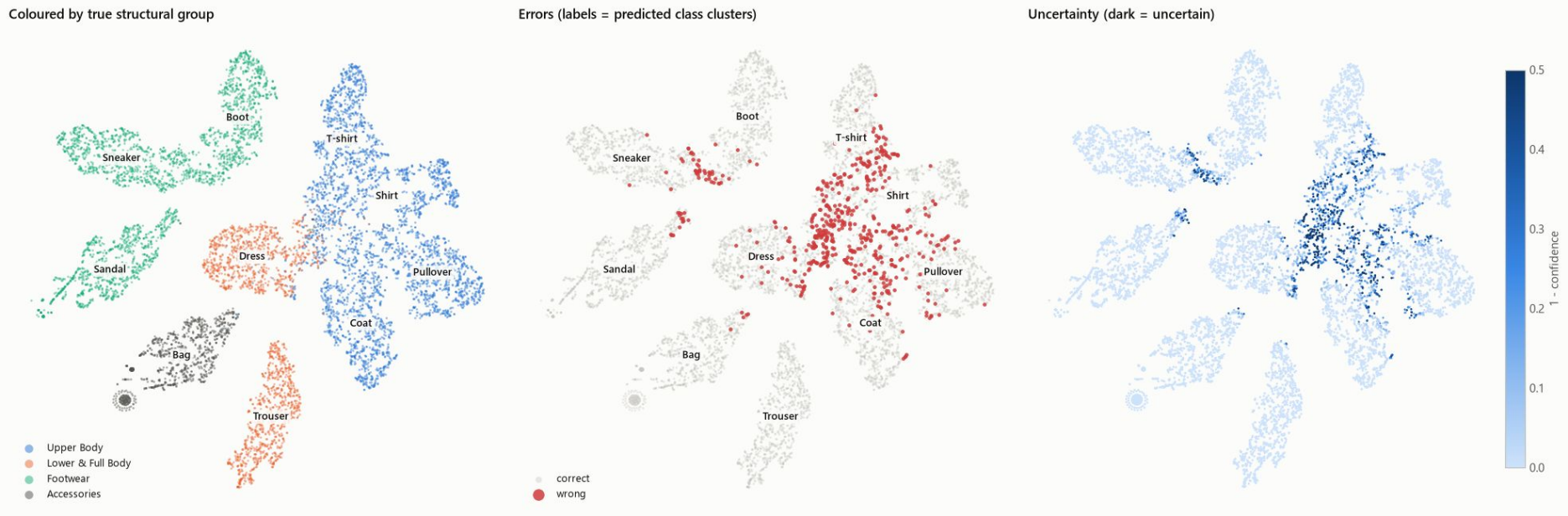
Vis4AI - t-SNE

Inference: through our t-SNE visualisation we can observe the predicted class clusters of our test dataset. Here the overlapping clusters show the commonly misclassified items

Eg: the model is confused when classifying T-shirts and pullovers, sandals and sneakers



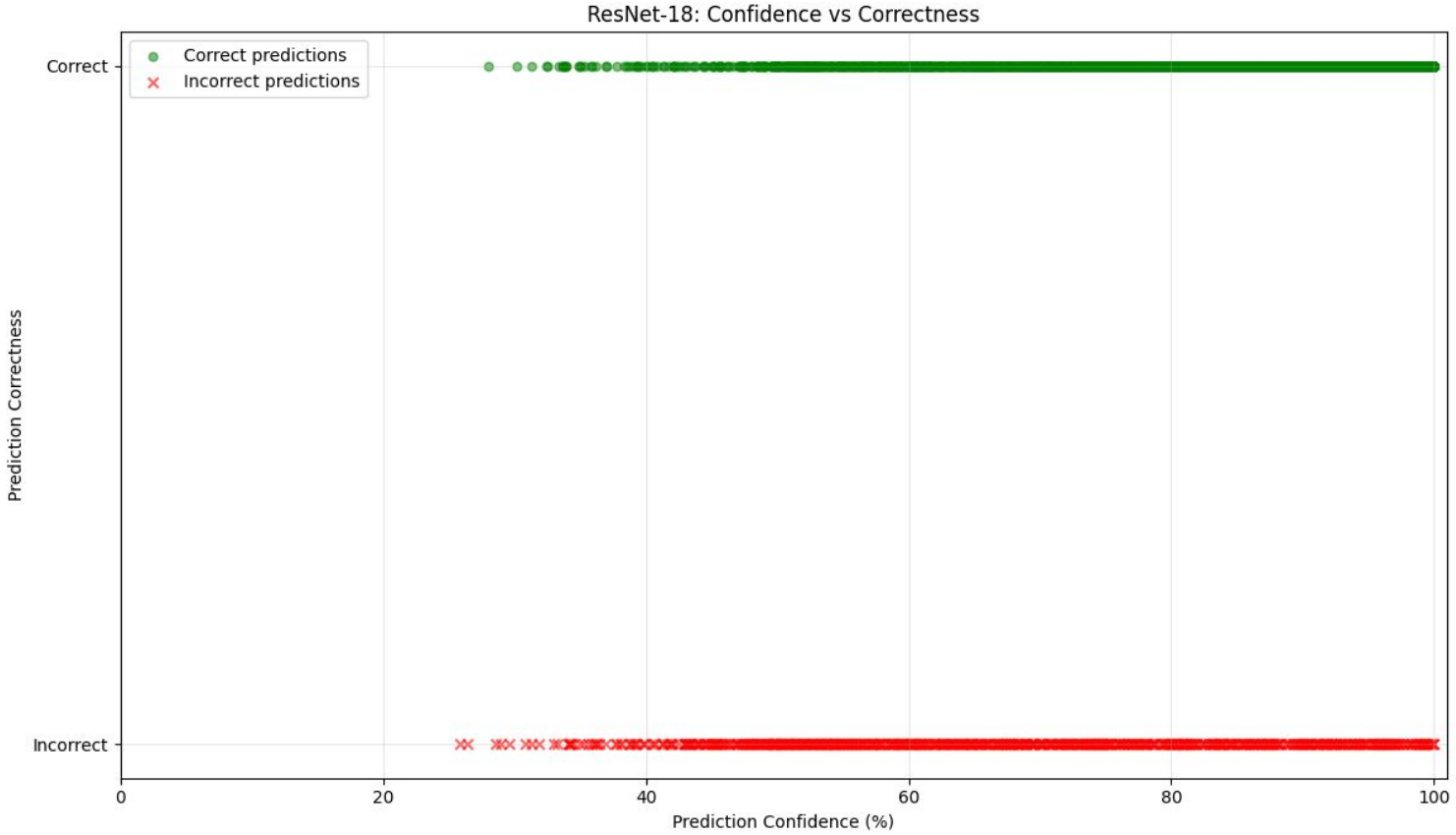
AI4Vis - Map of the output space



Inference: 26.1% of the errors are **camouflaged**: at least 8 of their 10 nearest neighbours in output space are correctly classified images of the predicted class; only the input view can explain them

Vis4AI - Confidence vs Correctness

Inference: it seems like the model is very confident even for incorrect predictions



Vis4AI - Misclassified examples

true_label	pred_label	
Shirt	Pullover	26
	T-shirt/top	16
Coat	Pullover	13
Sandal	Sneaker	13
Shirt	Coat	11
Ankle boot	Sneaker	8
Trouser	Dress	6
T-shirt/top	Shirt	5
Coat	Dress	5
Sneaker	Sandal	4
Shirt	Dress	3
Sneaker	Ankle boot	3
Ankle boot	Sandal	3
Dress	Trouser	3
Pullover	Dress	3
T-shirt/top	Bag	3
	Pullover	3
Coat	Shirt	2
Pullover	T-shirt/top	2
T-shirt/top	Dress	2
Ankle boot	Bag	1
Bag	Pullover	1
Ankle boot	Trouser	1
Bag	Sneaker	1
	Sandal	1
Pullover	Shirt	1
Coat	Trouser	1
Dress	Coat	1
	T-shirt/top	1
Shirt	Trouser	1
	Bag	1
T-shirt/top	Trouser	1
Trouser	Bag	1

	image_idx	true_label	pred_label	is_correct	max_confidence
23	23	Ankle boot	Sandal	False	0.994411
32	32	Dress	Trouser	False	0.910736
68	68	Ankle boot	Sneaker	False	0.910927
147	147	Shirt	Dress	False	0.946276
244	244	Shirt	Pullover	False	0.995785
382	382	Pullover	T-shirt/top	False	0.958705
634	634	Shirt	Coat	False	0.923064
701	701	Trouser	Dress	False	0.965071
722	722	Sandal	Sneaker	False	0.973347
787	787	Trouser	Dress	False	0.970660
909	909	Shirt	Pullover	False	0.903380
938	938	Shirt	Pullover	False	0.905102
979	979	Dress	T-shirt/top	False	0.989626
1000	1000	T-shirt/top	Shirt	False	0.971659
1051	1051	Sandal	Sneaker	False	0.943668
1077	1077	Coat	Pullover	False	0.957063
1111	1111	T-shirt/top	Dress	False	0.998386
1131	1131	Shirt	Coat	False	0.900058
1162	1162	Shirt	Coat	False	0.904664
1231	1231	T-shirt/top	Shirt	False	0.916879

AI4Vis - Error-flow network

Arrows show error flow from true class to prediction

Thicker = more errors.

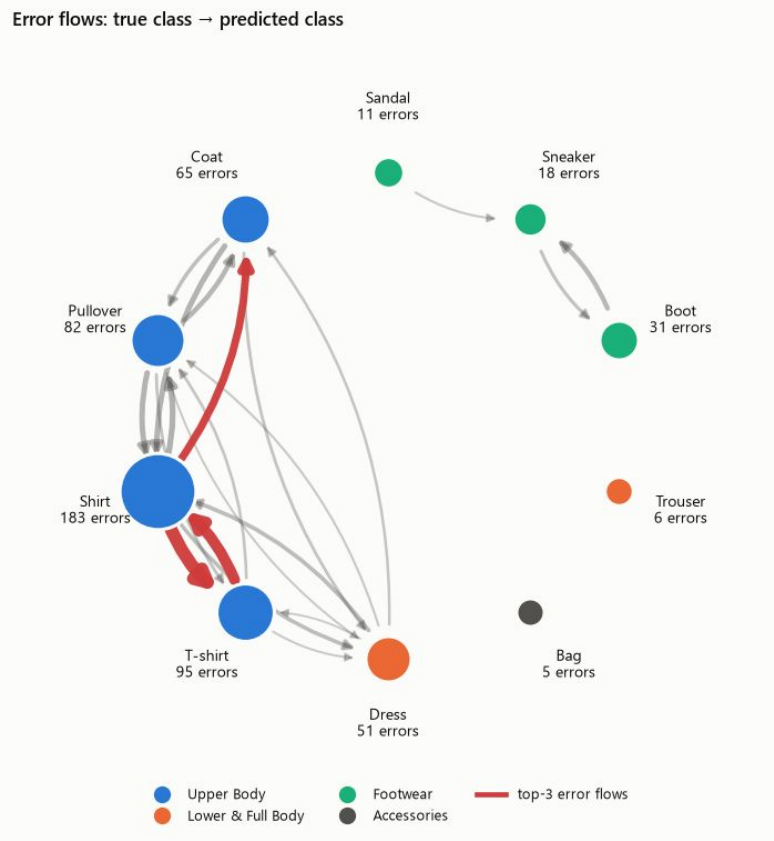
Shirt: 183 of 547 errors (33%).

5 largest flows are 48% of all errors, all between tops.

Footwear: errors stay within footwear.

Trousers and bags almost never fail.

The model mixes up garments that look alike, and in both directions.



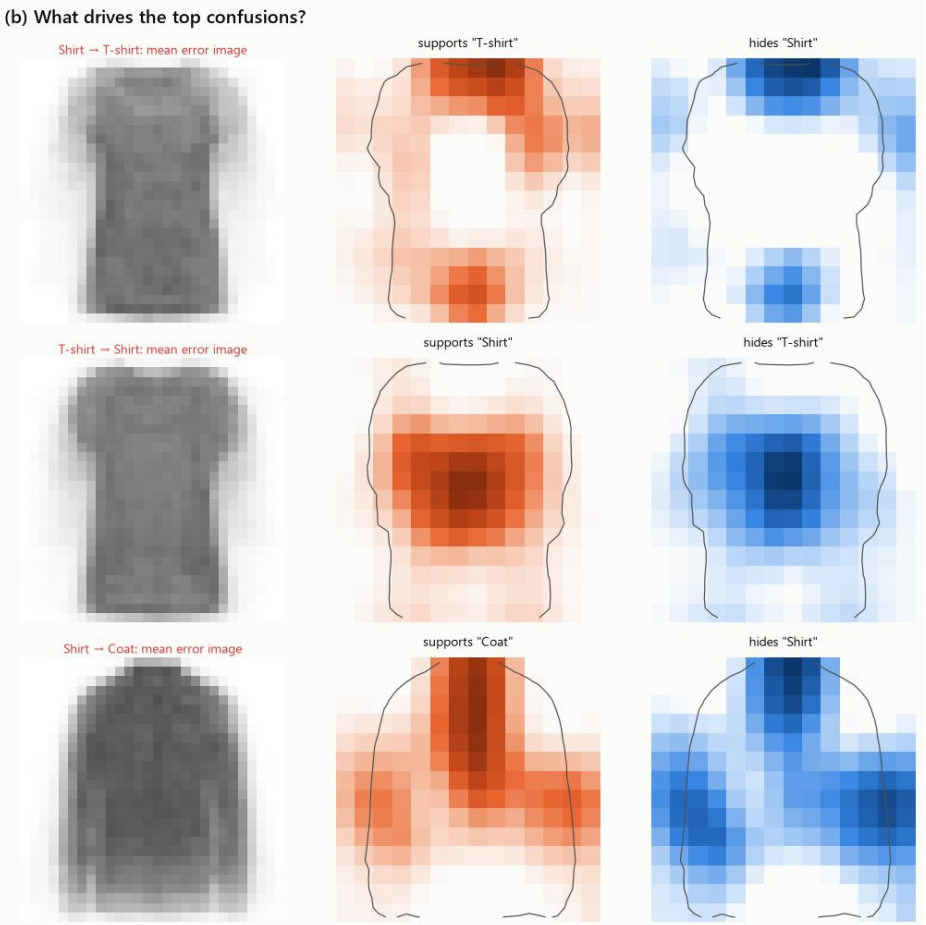
AI4Vis

Which regions decide the errors?

We erase one small patch (6x6 pixels) at a time and see how the models answer changes.

Orange region pushes model toward the wrong class.

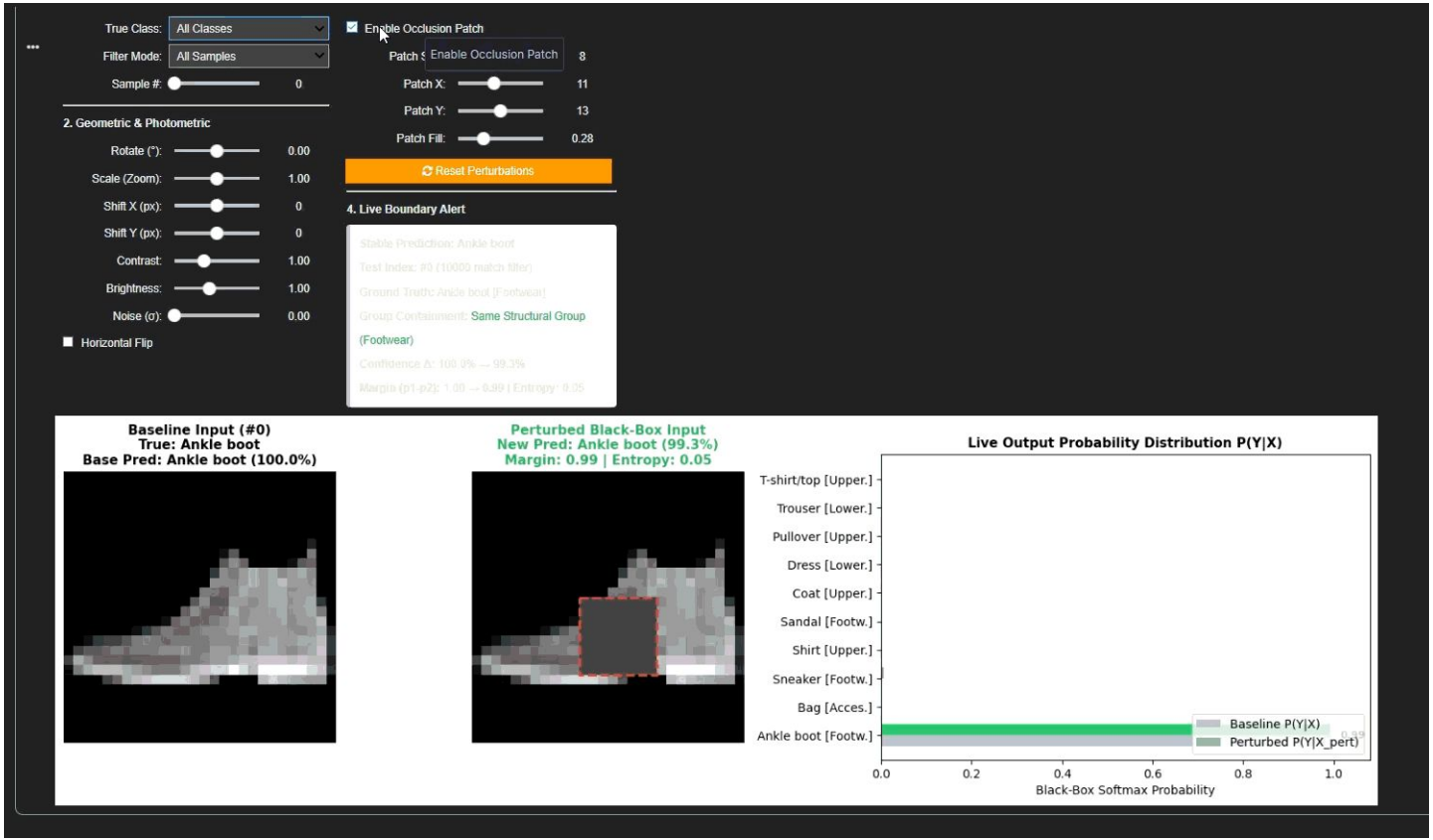
Blue region hides the true class.



AI4Vis

Shirt -> T-Shirt

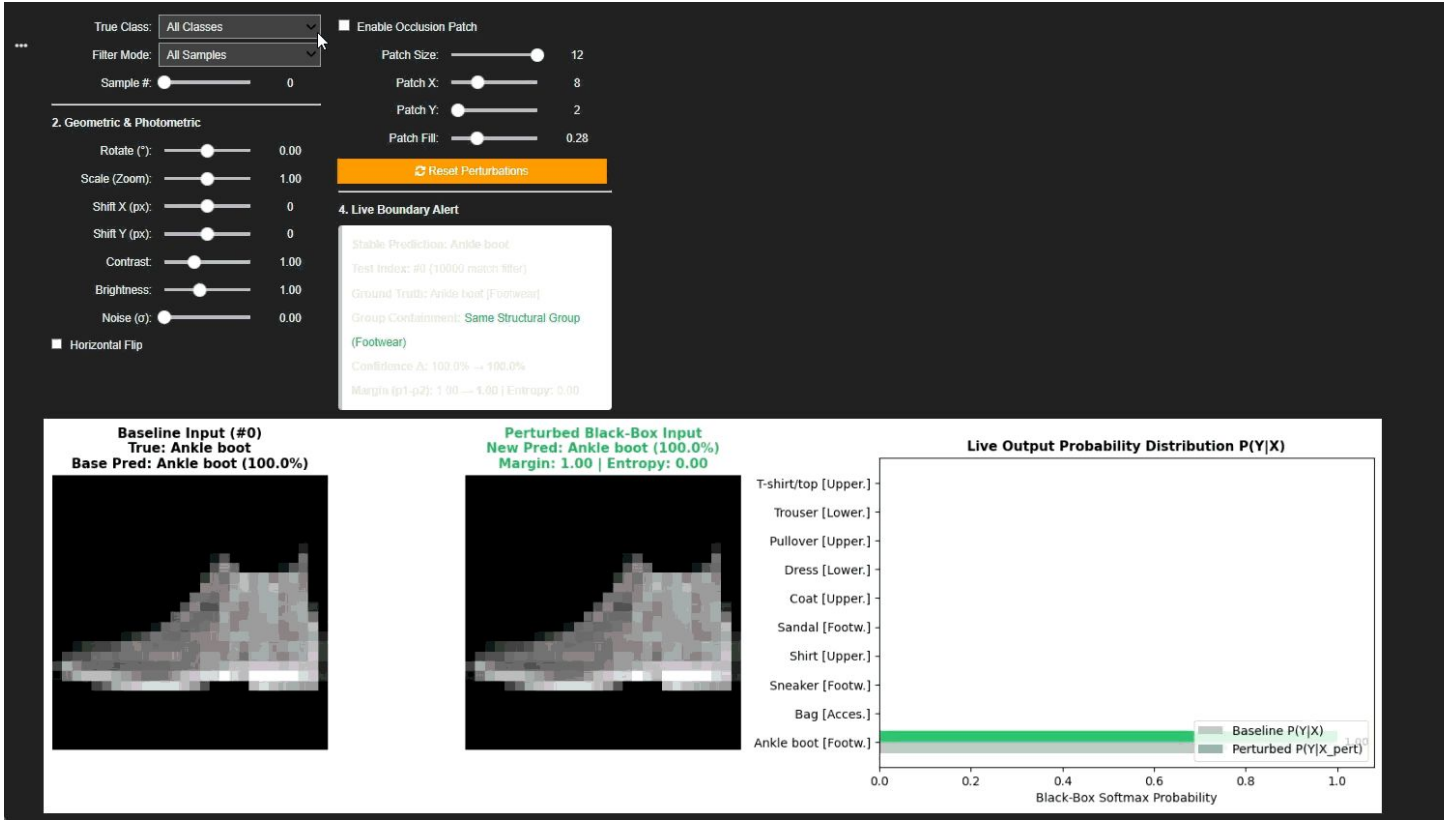
Hiding the **outer** structural confidence of Shirts



AI4Vis

T-Shirt -> Shirt

Hiding the **inner** structural confidence of T-Shirts



AI4Vis

Shirt <- Coat

Hiding the wrong confidence

